

REVIEW

EFFECTS OF POST-EXTRACTION ALVEOLAR RIDGE PRESERVATION VERSUS IMMEDIATE IMPLANT PLACEMENT: A SYSTEMATIC REVIEW AND META-ANALYSIS

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ABSTRACT

Objectives

This systematic review aimed to compare the clinical data including success rates, tissue preservation, esthetic results, and patient-reported outcomes between delayed implant placement after alveolar ridge preservation (ARP) and immediate implant placement (IIP).

Material and methods

Both electronic and manual searches were performed for randomized controlled trials and cohort studies consisting of at least 10 cases per group and a follow-up of at least 1-year in duration. The primary outcome was the implant success rate and secondary outcomes were changes in marginal bone level (MBL), pink esthetic score (PES) and patient reported outcomes consisting of complications and satisfaction.

Results

A total of 12 studies were included (8 randomized controlled trials and 4 cohort studies). This review contained 456 implants placed after ARP and 459 implants placed through IIP. The results from this meta-analysis showed that the success rates of implants placed through ARP protocol (98.68%) was significantly higher than that of implants placed through IIP protocol (95.21%) (RR = 1.03; 95% CI [1.01; 1.06]; $P = .008$; $I^2 = 0\%$).

Conclusion

The results from this meta-analysis and systematic review showed that implants placed through ARP protocol may demonstrate higher success rates compared to implants placed through IIP.

INTRODUCTION

One of the main prerequisites of successful implant treatment is the availability and sufficient amount of bone surrounding the extraction socket. Upon extraction of a single tooth, the alveolar ridge begins to recede, which may lead to

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KEYWORDS

Alveolar ridge augmentation, Dental implants, Immediate dental implant loading

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less-than-ideal conditions for implant placement after a period of healing due to losses in soft tissue levels and both bone width and height. Investigations conducted over the past 2 decades¹⁻⁶ have confirmed these findings in clinical studies. Changes in both hard tissue dimensions and soft tissue may lead to inferior success rates, less-than-ideal esthetic results, and less satisfaction.^{7,8} In an attempt to alleviate the effects of post-extraction bone resorption, 2 main protocols, alveolar ridge preservation⁹ and immediate implant placement,¹⁰ have been proposed. Although both ARP and IIP protocols are chosen by dentists globally, a consensus has yet to be reached regarding the superiority of one protocol over the other.

Alveolar ridge preservation, first proposed as “bone maintenance” in 1982,¹¹ is a procedure conducted after a complete tooth extraction with the purpose of preserving the alveolar ridge to maximize the available hard and soft tissue needed for future implant procedures. This procedure commonly uses bone grafts (allogenic, xenogenic, or human) to fill the extraction socket as well as a barrier membrane (often collagen) to cover the graft material and preserve the bone for future implant placement. ARP has been promoted and has clinically proven its ability to minimize the resorption of the alveolar ridge and thus soft tissue in the process.¹²⁻¹⁴ A number of systematic reviews^{4,13,15-17} have credited the protocol with fewer biomechanical complications, limited bone resorption, and esthetics.

The current indications for ARP can be generally grouped into the following:

- 1 Sites where the buccal plate is less than 1.5-2 mm thick;
- 2 Sites where there has been damage or loss of one or more of the socket walls;
- 3 Sites where maintaining bone volume is crucial to minimize the risk of involving anatomical structures, such as the posterior maxilla or mandible, where the maxillary sinus or inferior alveolar nerve may present as a complication if further bone is lost;
- 4 Patients with high aesthetic demands, which is prone to more recession;
- 5 Patients where many teeth are to be extracted and preservation of the bone is important for further restoration.¹⁸

Alternatively, immediate implant placement, first dated in 1978,¹⁹ is a procedure conducted when an implant is immediately placed into the fresh socket after tooth extraction. IIP has been supported by dentists for its similar clinical results.²⁰⁻²³ Immediate placement may also provide a shorter treatment period, comparable esthetic results to ARP, and a possibility for immediate partial loading and function benefiting the patient.²⁴⁻²⁸

The current indications for IIP can be generally grouped into the following:

- 1 Traumatic loss of teeth with a small amount of bone loss;
- 2 Tooth lost because of gross decay without purulent exudates or cellulites;
- 3 Inability to complete endodontic therapy;
- 4 Previous severe periodontal bone loss without purulent exudates;
- 5 Adequate soft tissue health to obtain primary wound closure.^{29,30}

This systematic review aimed to compare implant success rates, tissue preservation, esthetics, complications, and patient-reported outcomes between implants placed after alveolar ridge preservation and immediate implant treatment. The authors hypothesize that implants placed after alveolar ridge preservation present higher success rates compared to immediately placed implants.

MATERIAL AND METHODS

PRISMA Statement

This systematic review was written and designed according to the 2020 PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) checklist.^{31,32}

Research Question

The authors would like to address the question: What are the different effects ARP presents compared to IIP regarding clinical and patient-reported outcomes?

A PICO strategy shown below was utilized for the construction of the research question.

PICO

Population: Systemically healthy adult patients requesting implant treatment

Intervention: ARP (alveolar ridge preservation) and further treatment

Comparison: IIP (immediate implant placement) and further treatment

Outcome:

Primary Outcome: Implant success rate

Secondary Outcome: Marginal bone level changes (radiographical), hard and soft tissue preservation, complications, esthetics (PES scores), and patient satisfaction

Search Strategy

An electronic search involving 3 databases (PubMed, Web of Science, and China National Knowledge Infrastructure

(CNKI) was restricted to the English language in PubMed and Web of Science, restricted to the Mandarin Chinese language in China National Knowledge Infrastructure, and a date of publication range from January 1, 2000, to August 27, 2021. This systematic review was restricted to clinical studies. Search strategy was conducted with both text words and medical subject headings (MeSH) terms shown in Table 1.

Moreover, manual searches in the following journals were performed up to August 2021 to include additional relevant information: *Journal of Clinical Periodontology*, *Periodontology 2000*, *Journal of Periodontology*, *European Journal of Oral Implantology*, *Journal of Prosthodontics*, *Clinical Implant Dentistry and Related Research*, *Journal of Oral and Maxillofacial Surgery*, *Clinical Oral Implants Research*, *International Journal of Oral and Maxillofacial Surgery*, *Journal of Evidence-Based Dental Practice*, *International Journal of Oral and Maxillofacial Implants*, *International Journal of Implant Dentistry*, *International Journal of Prosthodontics*, *Journal of Craniomaxillofacial Surgery*, *Clinical Oral Investigations*, and *Journal of Dentistry*, *Journal of Prosthetic Dentistry*.

Additionally, gray literature was searched to screen for relevant reviews that may have been missed during the initial database search. Furthermore, a screening of completed clinical trials was performed under National Institutes of Health Clinical Trials Research (ClinicalTrials.gov) for extra data collection.

Inclusion and Exclusion Criteria

Clinical studies were included when the following criteria were met:

- 1 Randomized controlled trials or cohort studies with a minimum of ten patients allotted in each group
- 2 Patients were adults (18+) and systemically healthy
- 3 Follow-up period was at least 12 months
- 4 Trials comparing ARP and IIP directly
- 5 Trials reporting outcomes for implant success and any outcomes necessary for this study

References were excluded if any of the following criteria were met:

- 1 Studies did not include ARP or IIP as a comparison group
- 2 Follow-up period was less than 12 months
- 3 An updated version of the study was available
- 4 Patients were edentulous
- 5 Case reports and case series
- 6 Non-human subjects

Selection Process

Results of both the electronic and manual searches were screened by 2 authors (X.Y. and F.T.) independently accord-

Table 1. Search strategy.	
Database	Search strategy
China National Knowledge Infrastructure (CNKI)	ridge preservation AND immediate implant
Web of Science	alveolar ridge preservation (Topic) or ridge preservation (Topic) or socket preservation (Topic) and immediate implant (Topic) or immediate placement (Topic) or fresh socket (Topic) or bone graft (Topic) and human (Topic).
PubMed	(((((alveolar ridge preservation) OR alveolar bone preservation) OR alveolar ridge augmentation) OR alveolar bone augmentation) OR bone substitutes) OR socket preservation) OR bone graft) OR preserved sites) OR guided bone regeneration))) AND (((immediate implant) OR immediate dental implant) OR immediate placement) OR fresh socket).

ing to the inclusion and exclusion criteria listed above. Differences in opinion were settled through discussion with the third reviewer (D.Y.) being consulted when discussions were inconclusive. Subsequently, both the title and abstract of the studies selected were independently and manually screened for relevancy and full text was retrieved when information was lacking. Afterwards, studies from the final selection were subject to full-text screening based on the inclusion and exclusion criteria before relevant studies were selected for this study. Final selection was performed under the consensus of both authors (X.Y. and F.T.). Studies that failed to meet the criteria were excluded with reasons provided. Furthermore, in the case of duplicate references, the most recently published study was chosen.

Synthesis Method

Relevant data were selected using digital data table sheets by 2 independent reviewers (X.Y. and F.T.). Discord was solved through discussion with the third reviewer (D.Y.) being consulted when a consensus was not reached. In the case of unclear results presented in a study, an attempt to contact the author was conducted.

The following information was obtained:

Author, year of publication, type of study, patients included, follow-up period, treatment protocol (ARP and IIP), mean age, number of implants placed, socket condition, smoking habits, bone grafting and membrane materials, flap procedure, implant success rate, changes in marginal bone level (MBL), pink esthetic score (PES), complications, and patient-reported outcomes.

Quality Assessment

Two reviewers (X.Y. and F.T.) independently assessed the randomized controlled trials and cohort studies using Version 1 of the Cochrane risk-of-bias tool (RoB 1) for randomized trials³³ and the Newcastle-Ottawa Scale for cohort studies,³⁴ respectively.

For this study, a low risk of bias was concluded when the study met all the criteria in the “low risk of bias” parameter, a moderate risk of bias was concluded when the study was inconclusive in one of the parameters, and a high risk of bias was concluded when the study was inconclusive in 2 or more of the parameters.

For nonrandomized studies, the Newcastle-Ottawa Scale final scores were interpreted on a 0-9 scale, with 0-3 considered low quality, 4-6 considered moderate quality, and 7-9 considered high quality.

Statistical Analysis

For the quantitative analyses, relevant data obtained from the final studies selection were analyzed by performing both a Cochran's Q test and the DerSimonian and Laird method³⁵

for the I^2 value using the Review Manager (RevMan) 5.4.1 software to evaluate statistical heterogeneity among the studies.

For continuous variables (eg, MBL and PES), mean and standard deviation (SD) were utilized and recorded as mean differences. For dichotomous variables (eg, implant success rates), the risk ratio and accompanying 95% confidence intervals were calculated according to the number of events and displayed.

Analyses of the data were proceeded with different models depending on the P value of the Q test and I^2 value. The fixed-effect model was chosen when the study presented a P value greater than .05 and the I^2 value ranging from 0%-40%. However, when the study presented a P value greater than .05 and the I^2 value greater than 40%, the random-effect model was chosen instead.

A set of subgroup analyses were performed to investigate potential differences in results due to posterior or anterior implant placement, placement choice in the maxilla or mandible, and trial design.

Furthermore, a funnel plot was produced using the aforementioned Review Manager 5.4.1 software to address possible publication bias and other biases regarding implant success.

RESULTS

Study Selection

After the final selection concluded, a total of twelve studies³⁶⁻⁴⁷ were included. A total of 15 studies^{2,8,22,23,27,48-57} were excluded with reasons provided in Supplementary Table 1. The PRISMA flowchart developed according to this review is provided in Figure 1. Search for completed clinical trials with results were conclusive, with 3 registered studies that mentioned both ARP and IIP; however, all 3 trials were not incorporated into the final selection either due to a lack of comparison between ARP and IIP groups (ClinicalTrials.gov Identifier: NCT01794806) or because the study did not address the primary outcome designated for this review (ClinicalTrials.gov Identifier: NCT03690973; NCT03422458). Upon completing a scan of the electronic databases, gray literature, and ClinicalTrials.gov, no further studies were identified.

Description of Included Studies

Twelve studies that met the inclusion criteria were included in this meta-analysis. Eight studies were randomized controlled trials^{36,37,39-41,43,44,47} and 4 studies were cohort studies.^{38,42,45,46} The follow-up periods post-loading ranged from 1-5 years. Detailed descriptions of the selected studies are presented in Tables 2 and 3.

Table 2. Characteristics of the included studies.

Author 1 et al. Year	Type of Study	Patients Included	Follow-up (mo)	Treatment protocol	Mean age (y)	Implant number	Implant placement	Socket condition	Smoking (Cigarettes/d: No. of participants)	Key results
Slagter et al. 2021 ³⁶	RCT	40	60	IIP ARP	44 49	20 20	Maxillary esthetic zone (non-molar)	Buccal bony defects ≥ 5 mm	None	Mean marginal bone level change was not significant between ARP and IIP
Xu et al. 2017 ³⁸	Cohort study	127	12	IIP ARP	31.2 37.4	59 68	Maxillary esthetic zone (non-molar)	Buccal wall intact	Not detailed but participants $>20/d$ were not allowed	Peri-implant bone loss was not significant between ARP and IIP
Tonetti et al. 2017 ³⁹	RCT	124	36	IIP ARP	50 55	62 62	Maxillary or mandibular anterior (non-molar)	Not determined	Less than 20/d: 9 Less than 20/d: 9	IIP showed significantly lower aesthetic scores
Tallarico et al. 2017 ⁴⁰	RCT	24	12	IIP ARP	51.6 56.2	12 12	Maxillary or mandibular molar	Buccal wall intact	None	ARP was associated with less marginal bone loss and better aesthetic outcome
Esposito et al. 2017 ³⁷	RCT	140	12	IIP ARP	55.3 55.8	70 70	Maxillary esthetic zone (non-molar)	Well preserved, partially preserved, and poorly preserved buccal bone	Up to 10/d: 18; 10+/d: 4; Non-smokers: 48 Up to 10/d: 21; 10+/d: 7; Non-smokers: 42	Mean peri-implant marginal bone loss was not significant between ARP and IIP
Checchi et al. 2017 ⁴¹	RCT	100	12	IIP ARP	52.9 54.4	54 53	Maxillary or mandibular molar	No vertical loss of the buccal bone in relation to the palatal wall	Up to 10/d: 10; 10+/d: 3; Non-smokers: 37 Up to 10/d: 7; 10+/d: 4; Non-smokers: 39	Marginal bone levels at implant insertion were significant, favoring ARP
Guarnieri et al. 2016 ⁴²	Cohort study	25	36	IIP ARP	42 40	12 13	Maxillary esthetic zone (non-molar)	Buccal wall intact	Not detailed but participants $\geq 20/d$ were not allowed	Survival rate and mean bone level differences were not significant between ARP and IIP

(continued on next page)

Table 2 (continued)

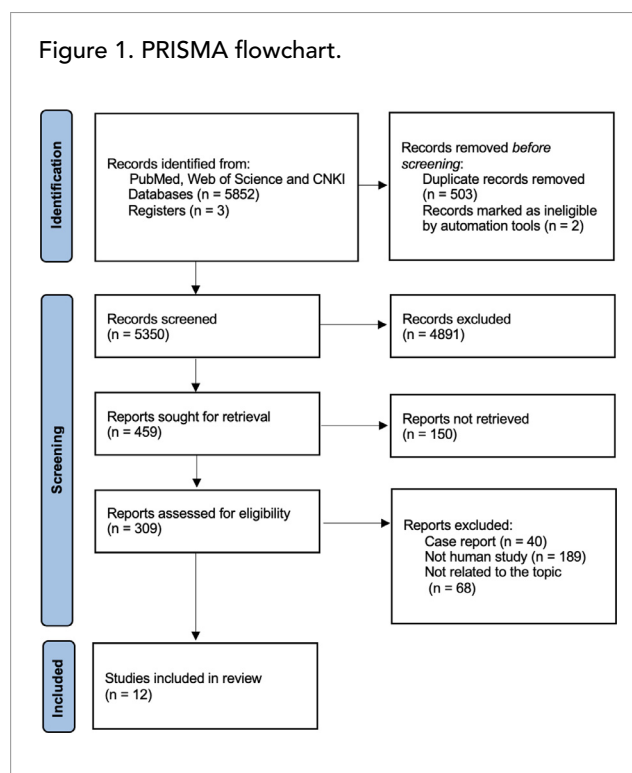
Author 1 et al. Year	Type of Study	Patients Included	Follow-up (mo)	Treatment protocol	Mean age (y)	Implant number	Implant placement	Socket condition	Smoking (Cigarettes/d: No. of participants)	Key results
Felice et al. 2015 ⁴³	RCT	50	12	IIP ARP	51.32 53.08	25 25	Maxillary esthetic zone (non-molar)	Less than 4 mm of buccal wall missing	Up to 10/d: 8; 10+/d: 3; Non-smokers: 14 Up to 10/d: 9; 10+/d: 0; Non-smokers: 16	Marginal bone levels at implant insertion showed significance; Mean aesthetic score showed no significance between ARP and IIP
Esposito et al. 2015 ⁴⁴	RCT	106	12	IIP ARP	48 50	54 52	Maxillary esthetic zone (non-molar)	Less than 4 mm of vertical loss of buccal bone in relation to the palatal wall	Up to 10/d: 8; 10+/d: 3; Non-smokers: 43 Up to 10/d: 7; 10+/d: 4; Non-smokers: 41	Marginal bone levels at implant insertion showed no significance; Mean aesthetic score also showed no significance
Grandi et al. 2013 ⁴⁵	Cohort study	50	12	IIP ARP	54.72 59.04	25 25	Maxillary esthetic zone (non-molar)	Buccal wall intact	Less than 20/d: 11 Less than 20/d: 8	Peri-implant bone resorption was similar in both groups; Ideal gingival marginal levels achieved in the ARP group
Cosyn et al. 2013 ⁴⁶	Cohort study	46	33 (IIP) 30 (ARP)	IIP ARP	51 53	28 18	Maxillary esthetic zone (non-molar)	Buccal bone wall intact Minor horizontal buccal bone defect (<1.5 mm)	Number of smokers: 5 Number of smokers: 4 Criteria not specified for both	Aesthetics scores were higher in the IIP group compared to the ARP group; Postoperative complications more common in the ARP group
Block et al. 2009 ⁴⁷	RCT	76	24	IIP ARP	Not Mentioned	38 38	Maxillary esthetic zone (non-molar)	Buccal wall intact	Not determined	IIP group preserved 1 mm more facial gingival margin position compared to the ARP group

ARP, alveolar ridge preservation; IIP, immediate implant placement; RCT, randomized clinical trial.

Table 3. Characteristics of the comparison and intervention groups.

Comparison (IIP)					Intervention (ARP)			
Author 1 et al. Year	Bone graft	Membrane type	Flap or flapless	Immediate loading	Bone graft	Membrane type	Flap or flapless	Immediate loading
Slagter et al. 2021 ³⁶	Xenograft and autograft	Soft tissue graft	Flapless	No	Xenograft and autograft	Soft tissue graft	Flap	No
Xu et al. 2017 ³⁸	Xenograft	Collagen (Bio-Gide)	Flapless when possible	No	Xenograft	Not determined	Flapless when possible	No
Tonetti et al. 2017 ³⁹	Xenograft	Collagen (Bio-Gide)	Flap	No	Xenograft	Collagen (Bio-Gide)	Flap	No
Tallarico et al. 2017 ⁴⁰	Xenograft	Not determined	Flapless	No	Xenograft	Collagen (BioHorizons)	Flap	No
Esposito et al. 2017 ³⁷	Xenograft	Equine pericardium derived collagen	Flap	No	Xenograft	Collagen (Evolution)	Flap	No
Checchi et al. 2017 ⁴¹	Autograft	Equine-origin collagen	Flapless when possible	No	Xenograft	Equine pericardium derived collagen	Flap	No
Guarnieri et al. 2016 ⁴²	None	Not determined	Flapless	Yes	Xenograft	Collagen (BioHorizons)	Flapless	Yes
Felice et al. 2015 ⁴³	Algae-derived bone substitute	Not determined	Flapless	Yes (16) No (9)	Algae-derived bone substitute	Collagen (Bio-Gide)	Flap	Yes (6) No (19)
Esposito et al. 2015 ⁴⁴	Xenograft	Collagen (Bio-Gide)	Flap	Yes (35) No (19)	Xenograft	Collagen (Bio-Gide)	Flap	Yes (13) No (39)
Grandi et al. 2013 ⁴⁵	Xenograft	Not determined	Flapless	Yes	Xenograft	Collagen	Flapless	Yes
Cosyn et al. 2013 ⁴⁶	Xenograft	Collagen (Bio-Gide)	Flap	Yes	Xenograft	Collagen (Bio-Gide)	Flap	No
Block et al. 2009 ⁴⁷	Allograft	None	Not determined	Yes	Allograft	None	Not determined	Yes

Figure 1. PRISMA flowchart.



The twelve included articles totaled 915 implants (456 in ARP protocol and 459 in IIP protocol) placed in 908 patients (453 in ARP protocol and 455 in IIP protocol). Smoking condition was reported in 11 studies with only Block et al.⁴⁷ lacking data, additionally, 3 studies did not provide specific data regarding participants' habits.^{38,42,46} Two publications contained strictly non-smoking patients.^{36,40} Four trials were conducted with 3 categorized groups (less than ten cigarettes per day, more than ten cigarettes per day, and non-smokers).^{37,41,43,44} Two studies provided the number of patients under 20 cigarettes per day with no further specifications.^{38,42} Nine studies reported implant placement only in the maxillary esthetic zone.^{36-38,42-47} Two trials designated only molar sites for implant placement.^{40,41} Tonetti et al.³⁹ contained implants placed in both the anterior maxilla and mandible.

Results for Risk of Bias and Quality Assessments

Of the 8 total randomized controlled trials displayed in Table 4, 6 trials were considered low risk of bias^{37,39-41,43,44} and 2 trials considered moderate risk of bias.^{36,47} Within the 4 non-randomized trials shown in Table 5, 2 studies scored an 8,^{38,42} 1 scored a 7,⁴⁶ 1 scored a 9,⁴⁵ and all categorized as high quality. Due to the inability to blind patients in this trial category, all studies that specified blinding of the dentist were considered low performance bias.

Categorical Descriptions for the Selected Studies

Socket condition

Six studies included buccal wall defect as an exclusion criterion,^{38,40-42,45,47} therefore, sockets with well-preserved buccal walls in the comparison and intervention groups were considered. Guarnieri et al.⁴² only set buccal wall defects as an exclusion criterion for the immediate placement group. Two studies accepted buccal bone defects of less than 4 mm.^{43,44} Esposito et al.³⁷ categorized sockets as well preserved, partially preserved, and poorly preserved, while Tonetti et al.³⁹ did not specifically state the condition of the sockets. On the other hand, Slagter et al.³⁶ strictly selected patients with a buccal wall defect of over 5 mm.

Implant placement

In the twelve selected studies, 9 studies involved implant placement in the maxillary esthetic zone, anterior maxilla, or second to second premolar as locations for implant placement,^{36-38,42-47} 3 studies inserted implants in the maxilla and mandible,³⁹⁻⁴¹ and 2 studies strictly placed implants in the maxillary and mandibular molars.^{40,41}

Implant Success Rate

In the twelve included studies, 887 out of 915 (96.94%) total implants were placed without failure, with 450 out of 456 (98.68%) implants placed without fail in the ARP group and 437 out of 459 (95.21%) implants placed without fail in the IIP group. Implant failure was defined by the study operators as implant mobility, removal of stable implants due to progressive marginal bone loss, and any mechanical complications that rendered the implant useless. Success rate for the ARP group was significantly higher when compared to the IIP group as shown in Figure 2 ($n=12$; $RR=1.03$; 95% CI [1.01; 1.06]; $P=.008$; $I^2=0\%$). Seven studies reported no implant failure in ARP groups^{36,38-40,42-44} while 4 studies reported no implant failure in IIP groups.^{36,38,40,42} Implant failure was primarily documented at the early stages (within 4 months) post-implant placement, as well as mostly occurring in the maxillary anterior. Block et al.⁴⁷ listed all failures during the 1 year follow up which was considered late-stage failure.

Subgroup analyses were performed to explore the influence of implant placement in different regions and trial design. Analytic results showed that implants placed in the molar (posterior) vs non-molar (anterior) sites were of no statistical significance between ARP and IIP groups ($n=2$; $RR=1.05$; 95% CI [0.96; 1.14]; $P=.28$; $I^2=0\%$). Success rate for ARP implants placed in the molar region was 96.92% compared to 92.42% in implants placed immediately. A comparison of studies that involved implants placed in the mandible yielded no statistical significance between ARP and IIP groups ($n=3$; $RR=1.03$; 95% CI [0.98; 1.08]; $P=.64$; $I^2=0\%$). Success rate for mandibular implants placed through ARP was 98.43% compared to 95.31% in the IIP group. Signif-

Table 4. Risk of bias assessment for randomized clinical trials (RCTs).

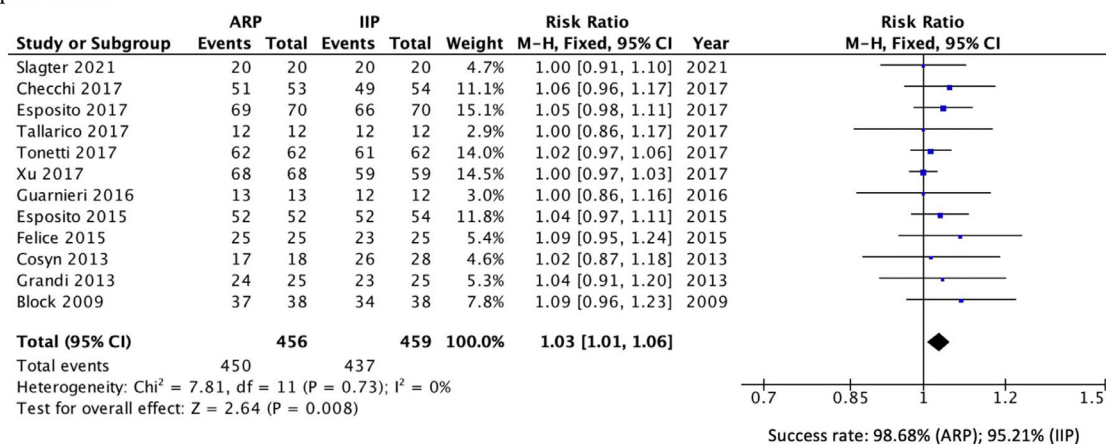
Criteria ³³	Slagter et al. 2021 ³⁶	Tonetti et al. 2017 ³⁹	Tallarico et al. 2017 ⁴⁰	Esposito et al. 2017 ³⁷	Checchi et al. 2017 ⁴¹	Felice et al. 2015 ⁴³	Esposito et al. 2015 ⁴⁴	Block et al. 2009 ⁴⁷
Random sequence allocation	L	L	L	L	L	L	L	L
Concealed allocation sequence	L	L	L	L	L	L	L	U
Performance bias	L	L	L	L	L	L	L	L
Detection bias	U	L	L	L	L	L	L	L
Attrition bias	L	L	L	L	L	L	L	L
Reporting bias	L	L	L	L	L	L	L	L
Risk of bias judgement	Moderate	Low	Low	Low	Low	Low	Low	Moderate

'Low' was given under performance bias.

H, high risk of bias; L, low risk of bias; SC, some concerns; U, unclear risk of bias in the case of blinded dentists (assessors) and unspecified patient blinding.

Figure 2. Meta-analysis for the comparison of implant success rates between different implant protocols. ARP, alveolar ridge preservation; CI, confidence interval; IIP, immediate implant placement; M-H, Mantel-Haenszel test.

Meta-analysis for the comparison of implant success rates between different implant protocols. CI, confidence interval; M-H, Mantel-Haenszel test; ARP, alveolar ridge preservation; IIP, immediate implant placement.



ificance in trial design was detected between ARP and IIP groups, favoring ARP, when only randomized controlled trials were included ($n = 8$; $RR = 1.04$; 95% CI [1.01; 1.08]; $P = .85$; $I^2 = 0\%$). Success rate for implants placed in the ARP group RCTs was 98.79% compared to 94.63% in the IIP group.

A funnel plot (Figure 5) for the implant success rate was provided to investigate potential publication bias. Visual symmetry in the diagram indicates a low risk of publication bias.

Marginal Bone Level

Radiographic data on marginal bone levels were reported in 9 studies^{36-38,40-45} and 3 studies lacked specific data or standard deviation^{39,46,47} and efforts to find the relevant data were inconclusive.

In all included studies, marginal bone level loss was measured from the implant collar to the first coronal point of visible bone-to-implant contact. All studies utilized a standard procedure that involved a periapical radiograph evaluation of peri-implant marginal bone levels.

Statistical analysis of the included studies displayed no significance in marginal bone level change between implant protocols as shown in Figure 3 ($n = 9$; Mean difference = -0.05 mm; 95% CI [-0.15; 0.05]; $P = .33$; $I^2 = 90\%$). The mean marginal bone level changes were 0.37 mm with SD 0.21 and 0.44 mm with SD 0.24 for the ARP and IIP groups, respectively.

Mid-Facial Mucosal Level

Statistical data regarding mid-facial mucosal level was presented in 2 studies that involved implant placement in the maxillary esthetic zone.^{36,42} Analysis of the data showed no significance between implant protocols ($n = 2$; Mean difference = -0.11 mm; 95% CI [-0.35; 0.14]; $P = .40$; $I^2 = 0\%$).

Pink Esthetic Score

Due to the high heterogeneity, a subgroup analysis was conducted for region of implant placement and trial design. In 2 studies that placed implants strictly in the molar region,^{40,41} significance was shown ($n = 2$; Mean difference = 1.36; 95% CI [0.65; 2.07]; $P = .0002$; $I^2 = 0\%$). When comparing randomized controlled trials with cohort studies, significance was not detected when strictly RCTs were included ($n = 6$; Mean difference = -0.12; 95% CI [-0.54; 0.79]; $P = .72$; $I^2 = 82\%$).

Statistical results showed no significance between ARP and IIP groups as shown in Figure 4 ($n = 8$; Mean difference = 0.10; 95% CI [-0.51; 0.72]; $P = .74$; $I^2 = 84\%$). The mean pink esthetic scores were 11.11 with SD 1.29 and 10.99 with SD 1.63 for the ARP and IIP groups, respectively.

Patient-Reported Outcomes

Complications

Implant complications were overall well documented in the included studies, with ten studies that described complications, 6 studies reported forms of biomechanical complications,^{37,39,41,43,44,46} and 4 studies reported no biomechanical complications of any kind.^{36,40,42,45} Of the 6 studies that mentioned forms of biomechanical complications, the most

Figure 3. Meta-analysis for the comparison of marginal bone level changes between different implant protocols. ARP, alveolar ridge preservation; CI, confidence interval; IIP, immediate implant placement; IV, inverse variance; SD, standard deviation.

Meta-analysis for the comparison of marginal bone level changes between different implant protocols. CI, confidence interval; IV, inverse variance; SD, standard deviation; ARP, alveolar ridge preservation; IIP, immediate implant placement.

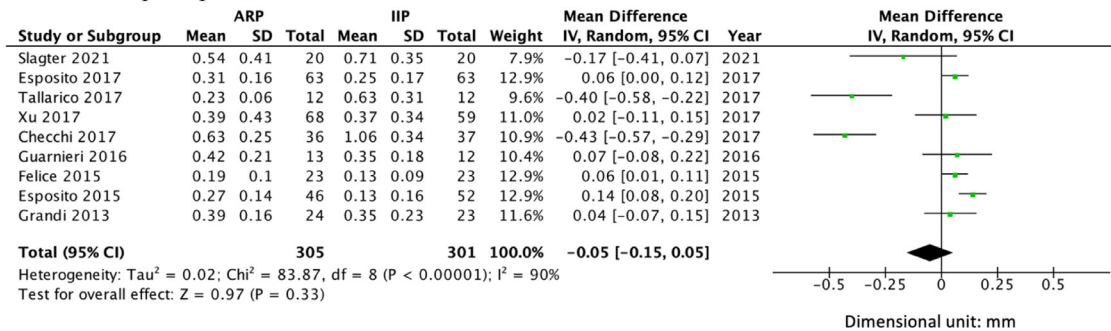
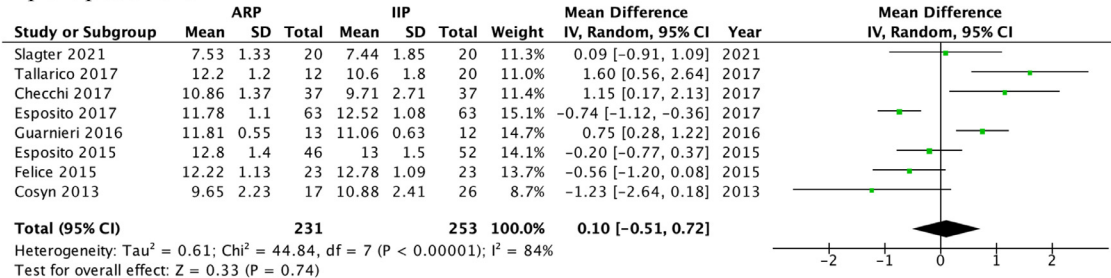


Figure 4. Meta-analysis for the comparison of pink esthetic scores between different implant protocols. ARP, alveolar ridge preservation; CI, confidence interval; IIP, immediate implant placement; IV, inverse variance; SD, standard deviation.

Meta-analysis for the comparison of pink esthetic scores between different implant protocols. CI, confidence interval; IV, inverse variance; SD, standard deviation; ARP, alveolar ridge preservation; IIP, immediate implant placement.



commonly reported were pain, discomfort, loosening, and fractures. This may show that the amount of alveolar bone available and the quality of said bone is an indispensable factor towards implant success. This opinion is supported by Friberg et al.,⁵⁸ where 25% (5 of 20) of complications recorded were a direct result due to bone deficiency. In contrast, a study conducted by Cosyn et al.⁴⁶ recorded 11 complications out of 18 total cases in the ARP group compared to only 5 out of 28 in the IIP group, the vast difference of this study compared to other studies may have been affected by the incorporation of a flap procedure. Another factor worth noting is the utilization of wide-diameter implants in studies conducted by Tallarico et al.⁴⁰ and Checchi et al.⁴¹ with the latter resulting in over twice the number of reported com-

plications in the IIP treatment group compared to the ARP group. This result may have occurred due to IIP requiring higher standards in the buccal bone condition and wide-diameter implants needing more socket space, which may have led to an inability to meet both specifications. However, since only 2 studies reported the use of wide-diameter implants, a small sample size is unable to represent a strong correlation between the 2 variables, therefore results should be interpreted with caution. Two studies did not provide specific data regarding technical complications,^{38,47} thus were excluded from the statistical analysis.

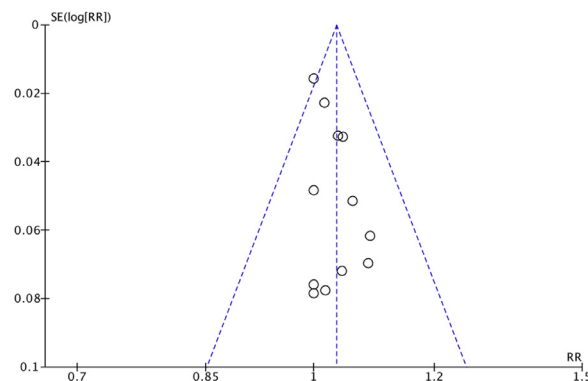
Significance regarding complications occurring was detected between ARP and IIP protocols, favoring ARP ($n = 4$; $RR = 0.43$; 95% CI [0.22; 0.83]; $P = .01$; $I^2 = 0\%$). Complica-

Table 5. Quality assessment utilizing the Newcastle Ottawa scale for cohort studies.

Author Year	Representativeness of the exposed cohort	Selection of the non-exposed cohort	Ascertainment of exposure	Demonstration that outcome of interest was not present at start of study	Comparability of cohorts on the basis of the design or analysis	Assessment of outcome	Was follow-up long enough for outcomes to occur	Adequacy of follow up of cohorts	Score
Xu et al. 2017 ³⁸	*	*	*	*	**	*	*	-	8
Guarnieri et al. 2016 ⁴²	*	*	*	*	-*	*	*	*	8
Grandi et al. 2013 ⁴⁵	*	*	*	*	**	*	*	*	9
Cosyn et al. 2013 ⁴⁶	*	*	*	*	-*	*	-	*	7

A (*) was given if the reference achieved the requirements in the 'Selection' and 'Outcome' categories, if none of the criteria were met, a (-) was given. A maximum of two (*) can be given in the 'Comparability' category, the first (*) given for alveolar process condition, the second (*) given for other controlled factors.

Figure 5. Funnel plot for implant success rate. RR, relative risk; SE(log[RR]), standard error.



tions were recorded in 11 out of 189 cases (5.82%) after ARP and 27 out of 196 cases (13.78%) after IIP. The most reported complications included pain, discomfort, mobility, and loosening.

Wide-diameter implants were chosen in 2 studies,^{40,41} including varying amounts of complications reported, with Checchi et al.⁴¹ reporting over double the number of complications in implants placed immediately compared to implants after ARP.

Patient satisfaction

Seven studies reported patient satisfaction, of which 4 studies used a questionnaire,^{37,41,43,44} 2 studies used a visual analogue scale (VAS),^{36,39} and one study used both a questionnaire and the VAS.⁴⁶ Four studies^{37,41,43,44} used questionnaires that designated 5 questions: "yes absolutely", "yes partly", "not sure", "not really", and "absolutely not". Slagter et al.³⁶ set the VAS scores ranging from 0-10, with 0 being completely dissatisfied and ten being completely satisfied. Tonetti et al.³⁹ also set the VAS scores ranging from 0-10, but with 0 representing no pain or discomfort whatsoever and ten representing extreme pain or discomfort. The results for both methods were used to determine if the patient was fully satisfied or only partially satisfied in most cases.

Upon analysis, all 7 studies reported no significance between treatment groups. Results suggested no significant difference in esthetic outcome and implant functionality, important factors in patient satisfaction. It is also worth mentioning that different studies have different focal points when it comes to esthetic evaluation. Moreover, it should be understood that patient interpretation of satisfaction may vary between respondents, thus leading to a lack of perception in esthetic differences. Nonetheless, an increase in number of

studies incorporating patient satisfaction is highly supported and favorable for future debate between the 2 protocols.

DISCUSSION

Consensus within the dental community has not been reached regarding the gold standard for post-extraction implant treatment protocol, with a number of systematic reviews^{15,16,28} published in the past 2 decades aiming for an answer. This lack of a gold standard for implant treatment protocols exists because of the previously mentioned indications different patients present which lead to specific treatment selections by dentists. On the other hand, patients may also choose their preferred treatment protocol based on factors including but not limited to financial capabilities, time availability, level of esthetic demand, etc. All of the aforementioned reasons may explain why the choice of treatment process is flexible rather than rigid.

Alveolar ridge preservation has gained popularity as a treatment choice and has been widely incorporated in the recent decade for site maintenance. This treatment protocol presents many advantages which include but are not limited to well preserved tissue conditions, sufficient bone integrity and availability, high success rates, and well-received esthetic outcomes.

Although ridge preservation has shown its ability to reduce bone dimensional resorption, immediate placement has gained attention for its shortened treatment time, similarly high success rates, and hastened loading capabilities. However, more prerequisites for immediate placement, including buccal wall integrity and completeness, exist and should be considered before proceeding, otherwise, a larger potential for compromised esthetics and implant failure may result. Moreover, Canellas et al.²⁸ concluded that a lack in achieving primary stability and additional bone formation in IIP are important factors affecting implant survival and esthetics. Furthermore, this review built upon the analyses from previous reviews^{15,16,28} by adding updated and additional RCT studies to the meta-analysis, introducing and comparing different study protocols (eg, cohort studies), and correcting known mistakes in the data from prior reviews.

The purpose of this meta-analysis and systematic review was to compare the effectiveness of ARP and IIP treatment protocols by factoring success rates, hard and soft tissue conditions, esthetic results, and patient-reported outcomes.

Results for Risk of Bias and Quality Assessments

Six trials were considered low risk of bias^{37,39-41,43,44} and 2 trials considered moderate risk of bias^{36,47} in the RCT studies. Reasoning for this designation comes from the conclusion of a manual assessment of the studies, where if the criteria was fulfilled "Low" was given, otherwise, "Unclear" or "High"

was given. However, in the case of blinded dentists (assessors) and unspecified patient blinding, "Low" was given under performance bias due to the necessary transparency between patients and dentists.

Two studies scored an 8,^{38,42} 1 scored a 7,⁴⁶ 1 scored a 9,⁴⁵ and all categorized as high quality in the cohort studies. Similar to the RCT studies, due to the necessary transparency between patients and dentists, all studies that specified blinding of the dentist were considered low performance bias.

Implant Success Rate

In line with conclusions from previous investigations,³⁶⁻⁴⁷ treatment after ARP (98.68%) boasts a significantly higher success rate compared to IIP (95.21%), as shown in this review. These results may not be representative due to the small sample size of cases available and the lack of studies conducted comparing the 2 treatment protocols.

Results of implant placement in the maxilla or mandible yielded no significance in success rate between 2 groups. This conclusion is upheld by Ketabi et al.,⁵⁹ where a success rate of 98% was reported in immediate molar (posterior) implantation, as well as no significance detected between maxillary and mandibular molar implants. On the other hand, Del Fabbro et al.⁶⁰ and Balaguer et al.⁶¹ both concluded that the difference in success rate for implants placed in the maxilla and mandible was statistically significant, favoring mandibular implant placement.

Clinicians should evaluate patient conditions thoroughly before deciding on a specific protocol.

Hard and Soft Tissue Preservation

Factors related to and influential towards hard and soft tissue preservation considered in this meta-analysis and systematic review include socket condition, smoking habits, choice of grafting material, and type of membrane covering.

Marginal bone level

In consensus with prior clinical trials and published reviews,^{36-38,40-45} loss in marginal bone levels were not significant between ARP and IIP treatment groups. The mean changes in marginal bone levels, however, were 0.37 mm in the ARP group and 0.44 mm in the IIP group, favoring ridge preservation. This result is supported by Atieh et al.,⁶² where changes in ridge dimensions were minimized when ARP was performed. On the other hand, some studies reported no significant diminishing of ridge height and width between ARP and IIP procedures, which shows the lack of consensus regarding this parameter.

Mid-facial mucosal level

In accordance with previously published reviews,^{36,42} comparisons of soft tissue preservation between the 2 treatment

groups resulted in no significance. This conclusion was also reinforced by this study, where no significance was determined, however, due to a small sample size ($n = 2$), bias in results should be carefully considered. On the other hand, van Nimwegen et al.⁶³ and a number of systematic reviews^{10,21} reckon that immediate implant placement bears significant risk for soft tissue recession due to bone resorption. The lack of measurements conducted regarding this outcome should be of concern because due to the degree of augmentation on the bone, different levels of ridge defects may appear which may lead to impaired esthetics.

Esthetic Results

Pink esthetic score

Results from the pink esthetic score analysis resulted in high heterogeneity, therefore, separate subgroup analyses were conducted. Pink esthetic scores of the molar region implants revealed a significance between treatment groups favoring ARP. Overall mean results showed that implants after ARP yielded a slightly higher PES. This result was in line with a number of studies that concluded that ridge preservation with bone grafting would result in better overall esthetic appearance.^{36,40-42} Contrary to certain studies, a portion of investigations concluded that IIP resulted in a significantly higher PES compared to ARP.^{37,43,44,46} This discrepancy in results reinforces the ongoing debate between treatment protocols.

LIMITATIONS

The limitations this systematic review presents should be taken into account, namely the heterogeneity of the marginal bone level changes and esthetics, the limited number of available studies conducted under each outcome, as well as the potential risk of bias. Moreover, this review only analyzed twelve studies and utilized 3 online databases, therefore, the results from this review should be interpreted with caution. Preferably, additional studies should be included, and more international databases and gray literature should be searched to allow for an ideal conclusion.

CONCLUSIONS

Based on the findings of this meta-analysis and systematic review, the following conclusions were drawn:

1. Implants placed after ARP have a significantly higher success rate than implants placed immediately.
2. Analysis of both hard (MBL) and soft (mid-facial mucosal level) tissue preservation demonstrated no significance between implant protocols.
3. Analysis of the esthetic outcome (PES) between both groups showed no significance.
4. Patient satisfaction was determined to be similarly high between ARP and IIP groups.

5. Further clinical trials with larger sample sizes should be conducted.

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DATA STATEMENT

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

SUPPLEMENTARY MATERIALS

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.jebdp.2022.101734](https://doi.org/10.1016/j.jebdp.2022.101734).

REFERENCES

1. Barone A, Ricci M, Tonelli P, Santini S, Covani U. Tissue changes of extraction sockets in humans: a comparison of spontaneous healing vs. ridge preservation with secondary soft tissue healing. *Clin Oral Implants Res*. 2013;24:1231–1237.
2. Garcia-Sanchez R, Mardas N, Buti J, Ortiz Ruiz AJ, Pardo Zamora G. Immediate implant placement in fresh alveolar sockets with a minimal split-thickness envelope flap: a randomised controlled clinical trial. *Clin Oral Implants Res*. 2021;32:1115–1126.
3. Atieh MA, Alsabeeha NH, Payne AG, Duncan W, Faggion CM, Esposito M. Interventions for replacing missing teeth: alveolar ridge preservation techniques for dental implant site development. *Cochrane Database Syst Rev*. 2015;CD010176. doi:10.1002/14651858.CD010176.pub3.
4. Garcia-Gonzalez S, Galve-Huertas A, Aboul-Hosn Centenero S, Mareque-Bueno S, Satorres-Nieto M, Hernandez-Alfaro F. Volumetric changes in alveolar ridge preservation with a compromised buccal wall: a systematic review and meta-analysis. *Med Oral Patol Oral Cir Bucal*. 2020;25:e565–e575.
5. Tonetti MS, Jung RE, Avila-Ortiz G, et al. Management of the extraction socket and timing of implant placement: consensus report and clinical recommendations of group 3 of the XV European workshop in periodontology. *J Clin Periodontol*. 2019;46(21):183–194 suppl.
6. Avila-Ortiz G, Chambrone L, Vignoletti F. Effect of alveolar ridge preservation interventions following tooth extraction: a systematic review and meta-analysis. *J Clin Periodontol*. 2019;46(21):195–223 suppl.
7. Cosyn J, De Lat L, Seyssens L, Doornewaard R, Deschepper E, Vervaeke S. The effectiveness of immediate implant placement for single tooth replacement compared to delayed implant

- placement: a systematic review and meta-analysis. *J Clin Periodontol.* 2019;46(21):224–241 suppl.
8. Mitsias M, Siompas K, Pistilli V, Trullenque-Eriksson A, Esposito M. Immediate, early (6 weeks) and delayed loading (3 months) of single, partial and full fixed implant supported prostheses: 1-year post-loading data from a multicentre randomised controlled trial. *Eur J Oral Implantol.* 2018;11:63–75.
 9. Jung RE, Ioannidis A, Hämmerle CHF, Thoma DS. Alveolar ridge preservation in the esthetic zone. *Periodontol 2000.* 2018;77:165–175.
 10. Buser D, Chappuis V, Belser UC, Chen S. Implant placement post extraction in esthetic single tooth sites: when immediate, when early, when late? *Periodontol 2000.* 2017;73:84–102.
 11. Ashman A, Bruins P. A new immediate hard tissue replacement (HTR)TM for bone in the oral cavity. *J Oral Implantol.* 1982;10:419–452.
 12. Horvath A, Mardas N, Mezzomo LA, Needleman IG, Donos N. Alveolar ridge preservation. A systematic review. *Clin Oral Investig.* 2013;17:341–363.
 13. De Risi V, Clementini M, Vittorini G, Mannocci A, De Sanctis M. Alveolar ridge preservation techniques: a systematic review and meta-analysis of histological and histomorphometrical data. *Clin Oral Implants Res.* 2015;26:50–68.
 14. Avila-Ortiz G, Elangovan S, Kramer KW, Blanchette D, Dawson DV. Effect of alveolar ridge preservation after tooth extraction: a systematic review and meta-analysis. *J Dent Res.* 2014;93:950–958.
 15. Zhou X, Yang J, Wu L, et al. Evaluation of the effect of implants placed in preserved sockets versus fresh sockets on tissue preservation and esthetics: a meta-analysis and systematic review. *J Evid Based Dent Pract.* 2019;19:101336.
 16. Mareque S, Castelo-Baz P, Lopez-Malla J, Blanco J, Nart J, Valles C. Clinical and esthetic outcomes of immediate implant placement compared to alveolar ridge preservation: a systematic review and meta-analysis. *Clin Oral Investig.* 2021;25:4735–4748.
 17. Mello CC, Lemos CAA, Verri FR, Dos Santos DM, Goiato MC, Pellizzer EP. Immediate implant placement into fresh extraction sockets versus delayed implants into healed sockets: a systematic review and meta-analysis. *Int J Oral Maxillofac Surg.* 2017;46:1162–1177.
 18. Darby I, Chen S, De Poi R. Ridge preservation: what is it and when should it be considered. *Aust Dent J.* 2008;53:11–21.
 19. Schulte W, Kleinekenscheidt H, Lindner K, Schareyka R. The Tübingen immediate implant in clinical studies. *Dtsch Zahnarztl Z.* 1978;33:348–359.
 20. Weigl P, Strangio A. The impact of immediately placed and restored single-tooth implants on hard and soft tissues in the anterior maxilla. *Eur J Oral Implantol.* 2016;9(1):S89–106 suppl.
 21. Chen ST, Buser D. Esthetic outcomes following immediate and early implant placement in the anterior maxilla—a systematic review. *Int J Oral Maxillofac Implants.* 2014;29:186–215 suppl.
 22. Cosyn J, Eghbali A, Hermans A, Vervaeke S, De Bruyn H, Cleymaet R. A 5-year prospective study on single immediate implants in the aesthetic zone. *J Clin Periodontol.* 2016;43:702–709.
 23. Seyssens L, Eghbali A, Cosyn J. A 10-year prospective study on single immediate implants. *J Clin Periodontol.* 2020;47:1248–1258.
 24. Zhou YM YX, Wang XJ. Short-term clinical outcomes of immediate implantation versus delayed implantation after socket preservation in maxillary molar area for patients with severe chronic periodontitis. *J Oral Sci Res.* 2017;33:78–82.
 25. Covani U, Bortolaia C, Barone A, Sbordone L. Bucco-lingual crestal bone changes after immediate and delayed implant placement. *J Periodontol.* 2004;75:1605–1612.
 26. Raes F, Cosyn J, De Bruyn H. Clinical, aesthetic, and patient-related outcome of immediately loaded single implants in the anterior maxilla: a prospective study in extraction sockets, healed ridges, and grafted sites. *Clin Implant Dent Relat Res.* 2013;15:819–835.
 27. Salman A, Thacker S, Rubin S, Dhingra A, Ioannidou E, Schincaglia GP. Immediate versus delayed loading of mandibular implant-retained overdentures: a 60-month follow-up of a randomized clinical trial. *J Clin Periodontol.* 2019;46:863–871.
 28. Canellas J, Medeiros PJD, Figueredo C, Fischer RG, Ritto FG. Which is the best choice after tooth extraction, immediate implant placement or delayed placement with alveolar ridge preservation? a systematic review and meta-analysis. *J Cranio-maxillofac Surg.* 2019;47:1793–1802.
 29. Ebenezer V, Balakrishnan K, Asir RV, Sragunar B. Immediate placement of endosseous implants into the extraction sockets. *J Pharm Bioallied Sci.* 2015;7:S234–S237.
 30. Block MSKJN. Placement of endosseous implants into tooth extraction sites. *J Oral Maxillofac Surg.* 1991;49:1269–1276.
 31. Page MJ, McKenzie JE, Bossuyt PM, et al. Updating guidance for reporting systematic reviews: development of the PRISMA 2020 statement. *J Clin Epidemiol.* 2021;134:103–112.
 32. Page MJ, McKenzie JE, Bossuyt PM, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *PLoS Med.* 2021;18:e1003583. doi:10.1371/journal.pmed.1003583.
 33. Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA, eds. *Cochrane Handbook for Systematic Reviews of Interventions Version 6.2 (Updated February 2021): Cochrane*; 2021 Available at <http://www.training.cochrane.org/handbook> Accessed September 4, 2021.
 34. GA Wells BS, D O'Connell, J Peterson, V Welch, M Losos, P Tugwell. The Newcastle-Ottawa Scale (NOS) for Assessing the Quality of Nonrandomised Studies in Meta-Analyses. Uni-

versity of Ottawa, Ottawa, Canada. Available at: http://www.ohri.ca/programs/clinical_epidemiology/oxford.asp. Accessed September 4, 2021.

35. DerSimonian R, Laird N. Meta-analysis in clinical trials. *Control Clin Trials*. 1986;7:177–188.
36. Slagter KW, Meijer HJA, Hentenaar DFM, Vissink A, Raghoobar GM. Immediate single-tooth implant placement with simultaneous bone augmentation versus delayed implant placement after alveolar ridge preservation in bony defect sites in the esthetic region: a 5-year randomized controlled trial. *J Periodontol*. 2021. doi:10.1002/JPER.20-0845.
37. Esposito M, Zucchelli G, Cannizzaro G, et al. Immediate, immediate-delayed (6 weeks) and delayed (4 months) post-extractive single implants: 1-year post-loading data from a randomised controlled trial. *Eur J Oral Implantol*. 2017;10:11–26.
38. Xu JM, Chen DL, Wang P, Tang Y, Liu L, Tang CB. A retrospective research of the rapeutic effect of immediate implant and delayed implant following alveolar ridge preservation after minimally invasive extraction in anterior area. *Stomatology*. 2017;37:325–330.
39. Tonetti MS, Cortellini P, Graziani F, et al. Immediate versus delayed implant placement after anterior single tooth extraction: the timing randomized controlled clinical trial. *J Clin Periodontol*. 2017;44:215–224.
40. Tallarico M, Xhanari E, Pisano M, Gatti F, Meloni SM. Molar replacement with 7 mm-wide diameter implants: to place the implant immediately or to wait 4 months after socket preservation? 1 year after loading results from a randomised controlled trial. *Eur J Oral Implantol*. 2017;10:169–178.
41. Checchi V, Felice P, Zucchelli G, et al. Wide diameter immediate post-extractive implants vs delayed placement of normal-diameter implants in preserved sockets in the molar region: 1-year post-loading outcome of a randomised controlled trial. *Eur J Oral Implantol*. 2017;10:263–278.
42. Guarnieri R, Belleggia F, Grande M. Immediate versus delayed treatment in the anterior maxilla using single implants with a laser-microtextured collar: 3-year results of a case series on hard- and oft-tissue response and esthetics. *J Prosthodont*. 2016;25:135–145.
43. Felice P, Pistilli R, Barausse C, Trullenque-Eriksson A, Esposito M. Immediate non-occlusal loading of immediate post-extractive versus delayed placement of single implants in preserved sockets of the anterior maxilla: 1-year post-loading outcome of a randomised controlled trial. *Eur J Oral Implantol*. 2015;8:361–372.
44. Esposito M, Barausse C, Pistilli R, et al. Immediate loading of post-extractive versus delayed placed single implants in the anterior maxilla: outcome of a pragmatic multicenter randomised controlled trial 1-year after loading. *Eur J Oral Implantol*. 2015;8:347–358.
45. Grandi T, Guazzi P, Samarani R, Grandi G. Immediate provisionalisation of single post-extractive implants versus implants placed in healed sites in the anterior maxilla: 1-year results from a multicentre controlled cohort study. *Eur J Oral Implantol*. 2013;6:285–295.
46. Cosyn J, Eghbali A, Hanselaer L, et al. Four modalities of single implant treatment in the anterior maxilla: a clinical, radiographic, and aesthetic evaluation. *Clin Implant Dent Relat Res*. 2013;15:517–530.
47. Block MS, Mercante DE, Lirette D, Mohamed W, Ryser M, Castellon P. Prospective evaluation of immediate and delayed provisional single tooth restorations. *J Oral Maxillofac Surg*. 2009;67:89–107.
48. Naji BM, Abdelsameea SS, Alqutaibi AY, Said Ahmed WM. Immediate dental implant placement with a horizontal gap more than two millimetres: a randomized clinical trial. *Int J Oral Maxillofac Surg*. 2021;50:683–690.
49. Wang J, Lerman G, Bittner N, Fan W, Lalla E, Papapanou PN. Immediate versus delayed temporization at posterior single implant sites: a randomized controlled trial. *J Clin Periodontol*. 2020;47:1281–1291.
50. Tallarico M, Xhanari E, Pisano M, De Riu G, Tullio A, Meloni SM. Single post-extractive ultra-wide 7 mm-diameter implants versus implants placed in molar healed sites after socket preservation for molar replacement: 6-month post-loading results from a randomised controlled trial. *Eur J Oral Implantol*. 2016;9:263–275.
51. Slagter KW, Meijer HJ, Bakker NA, Vissink A, Raghoobar GM. Immediate single-tooth implant placement in bony defects in the esthetic zone: a 1-year randomized controlled trial. *J Periodontol*. 2016;87:619–629.
52. Esposito M, Siormpas K, Mitsias M, Bechara S, Trullenque-Eriksson A, Pistilli R. Immediate, early (6 weeks) and delayed loading (3 months) of single implants: 4-month post-loading from a multicenter pragmatic randomised controlled trial. *Eur J Oral Implantol*. 2016;9:249–260.
53. Karaca C, Er N, Gulsahi A, Koseoglu OT. Alveolar ridge preservation with a free gingival graft in the anterior maxilla: volumetric evaluation in a randomized clinical trial. *Int J Oral Maxillofac Surg*. 2015;44:774–780.
54. Pellicer-Chover H, Penarrocha-Oltra D, Bagan L, Fichy-Fernandez AJ, Canullo L, Penarrocha-Diago M. Single-blind randomized clinical trial to evaluate clinical and radiological outcomes after one year of immediate versus delayed implant placement supporting full-arch prostheses. *Med Oral Patol Oral Cir Bucal*. 2014;19:e295–e301.
55. Jokstad A, Alkumru H. Immediate function on the day of surgery compared with a delayed implant loading process in the mandible: a randomized clinical trial over 5 years. *Clin Oral Implants Res*. 2014;25:1325–1335.
56. Felice P, Soardi E, Piattelli M, Pistilli R, Jacotti M, Esposito M. Immediate non-occlusal loading of immediate post-extractive versus delayed placement of single implants in preserved sockets of the anterior maxilla: 4-month post-loading results from a

- pragmatic multicentre randomised controlled trial. *Eur J Oral Implantol*. 2011;4:329–344.
57. van Kesteren CJ, Schoolfield J, West J, Oates T. A prospective randomized clinical study of changes in soft tissue position following immediate and delayed implant placement. *Int J Oral Maxillofac Implants*. 2010;25:562–570.
 58. Friberg B. Bone augmentation for single tooth implants: a review of the literature. *Eur J Oral Implantol*. 2016;9(1):S123–S134 suppl.
 59. Ketabi M, Deporter D, Atenafu EG. A systematic review of outcomes following immediate molar implant placement based on recently published studies. *Clin Implant Dent Relat Res*. 2016;18:1084–1094.
 60. Del Fabbro M, Testori T, Kekovic V, Goker F, Tumedei M, Wang HL. A systematic review of survival rates of osseointegrated implants in fully and partially edentulous patients following immediate loading. *J Clin Med*. 2019;8(12):2142 Published 2019 Dec 4. doi:10.3390/jcm8122142.
 61. Balaguer J, Ata-Ali J, Penarrocha-Oltra D, Garcia B, Penarrocha-Diago M. Long-term survival rates of implants supporting overdentures. *J Oral Implantol*. 2015;41:173–177.
 62. Atieh MA, Alsabeeha NH, Payne AG, Ali S, Faggion CMJ, Esposito M. Interventions for replacing missing teeth: alveolar ridge preservation techniques for dental implant site development. *Cochrane Database Syst Rev*. 2021;4:CD010176.
 63. van Nimwegen WG, Raghoobar GM, Zuiderveld EG, Jung RE, Meijer HJA, Muhlemann S. Immediate placement and provisionalization of implants in the aesthetic zone with or without a connective tissue graft: a 1-year randomized controlled trial and volumetric study. *Clin Oral Implants Res*. 2018;29:671–678.